

NIAH Retrieval Depth-Sweep Heatmaps Across Lengths (2K -> 256K) and Depths (10%, 50%, 90%)

Polar: Bounded (34.4% @256K)

	2K	4K	8K	16K	32K	64K	128K	256K
10%	99	79	67	52	66	57	41	30
50%	99	99	98	83	69	50	44	34
90%	98	98	97	84	68	56	40	39

NoPE: Dilution (0.6% @256K)

	2K	4K	8K	16K	32K	64K	128K	256K
10%	100	94	49	48	34	12	4	0
50%	99	84	49	48	34	14	6	1
90%	99	50	49	47	37	15	6	1

RoPE: Collapse (0.0% @256K)

	2K	4K	8K	16K	32K	64K	128K	256K
10%	81	37	23	1	10	0	0	0
50%	88	7	18	0	0	0	0	0
90%	96	30	11	0	0	0	0	0

Raven Native: (20.3% @256K)

	2K	4K	8K	16K	32K	64K	128K	256K
10%	31	28	28	24	22	20	20	19
50%	36	37	28	24	23	21	20	19
90%	56	45	36	29	29	24	24	23

Atma-Raven-Titans: (10.0% @256K)

	2K	4K	8K	16K	32K	64K	128K	256K
10%	30	30	26	22	18	13	8	4
50%	32	29	25	21	19	16	12	8
90%	67	36	30	28	27	25	21	18